



Rajarata University of Sri Lanka

Department of Computing

ICT1305 – Data Structures
Laboratory 02– Structures in C
Time: 120 Minutes

Objective:

To understand the concept of structures in C programming, how they are declared, initialized, and used to group different types of data under one name.

What is a Structure?

A structure is a **user-defined data type** in **C** that allows grouping **variables of different data types** under one name.

Think of it as a **template** or a **record** — like an ID card that contains name (string), age (int), height (float) etc.

Why Use Structures?

- To represent real-world entities (like **student**, **employee**, **book**).
- To store heterogeneous data.
- To improve code readability and data organization.
- To pass complex data between functions.

Structure Syntax

```
struct StructureName {  
    dataType member1;  
    dataType member2;  
    ...  
};
```

Example 1: Student Record

```
1  #include <stdio.h>
2  #include <string.h> // Required for strcpy
3
4  struct Student {
5      int id;
6      char name[50];
7      float marks;
8  };
9
10 int main() {
11     struct Student s1;
12
13     s1.id = 101;
14     strcpy(s1.name, "Alice");
15     s1.marks = 85.5;
16
17     printf("ID: %d\n", s1.id);
18     printf("Name: %s\n", s1.name);
19     printf("Marks: %.2f\n", s1.marks);
20
21     return 0;
22 }
```

Key Concepts

Concept	Description
<code>struct</code> keyword	Used to define a structure.
<code>struct Student s1;</code>	Declares a structure variable.
<code>.</code> operator	Accesses members of a structure.
<code>strcpy()</code>	Required to copy string into char array.
Structures are stack-allocated	By default, structure variables are stored on the stack.

Summary Points for Students.

- Structures group multiple data types under one name.
- Access structure members using the **dot .** operator.
- Use **strcpy()** to assign values to string members.
- Always include necessary header files (like **<string.h>** when using **strcpy()**).

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